

Long-term outcomes of behavior problems after epilepsy surgery in childhood

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Abstract Although over 50 % of youth with intractable epilepsy present with behavioral problems, the long-term prognosis after resective epilepsy surgery in childhood is unclear. This study evaluated such outcomes in a cohort of surgical and nonsurgical patients. Participants were 108 patients (71 underwent surgery) with childhood-onset intractable epilepsy; their mean age at follow-up was 19.9 (standard deviation 4.3) years. The parent-rated Child or Adult Behavior Checklist (CBCL/ABCL) was used to document behavior prior to surgery and again 4–11 years later, and at comparable time points in the nonsurgical group. We focus primarily on externalizing and related symptomology in evaluating the CBCL/ABCL Externalizing Behavior summary scale and the Somatic Complaints, Thought Problems, Attention Problems, Aggressive Behavior, Intrusive, and Rule-breaking Behavior syndrome scales. Both groups improved in the Externalizing Behavior summary scale and in domains reflecting Somatic Complaints, Thought Problems, and Attention Problems from baseline to follow-up. Surgical and nonsurgical patients did not differ on any domain, whereas seizure-free patients had fewer symptoms in almost all behavioral domains, compared to patients with seizures. Regression analyses revealed that the most

consistent predictor of improved behavior was greater behavior problems at baseline. Younger age at baseline was associated with improvements in Externalizing Behavior. The described long-term outcomes of behavior problems among patients with childhood-onset intractable epilepsy are encouraging, in that modest improvements were noted among all patient groups. Furthermore, seizure freedom, whether achieved through surgery or medication management, was associated with fewer behavioral problems in most domains.

Keywords Behavior checklist · Pediatrics · Children · Attention · Externalizing · Intractable epilepsy

Introduction

Beyond disabling seizures, the quality of life of children with epilepsy is further and significantly impacted by the high prevalence of cognitive, emotional, and behavioral comorbidities. Although such comorbidities may be a consequence of the underlying aberrant neural substrate, the unpredictable nature of seizures and their impact on psychosocial functioning may be a significant contributor. Behavioral problems are particularly high among children with epilepsy, being three to five times more prevalent when compared to children with other chronic medical conditions and healthy controls, respectively [1]. Among patients with medically refractory epilepsy, behavioral problems are evident in over 50 % of patients [2, 3], and as many as 83 % have a psychiatric disorder [4].

A few studies have systematically evaluated the progression of behavioral problems using standardized questionnaires following pediatric epilepsy surgery. In this report, behavioral problems refer to externalizing and

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related symptomology, including somatic complaints, thought problems, attention problems, aggressive behavior, and rule breaking behavior. Studies that have evaluated these behavioral problems in the short-term after epilepsy surgery (typically up to 2 years) report inconsistent results; some report no significant change in any domain [3, 5], whereas others report a few improvements in specific domains [2, 6–9]. When improvements are evident, they are associated with seizure control [2, 7], the passage of time [8, 9], or epilepsy-related variables [6, 8]. Only one study has evaluated the long-term outcomes of pediatric epilepsy surgery. Hannan et al. [10] followed 11 surgical patients for up to 8.5 years and found improvements on the Total Difficulties score of the Strengths and Difficulties questionnaire in the short-term that were maintained at long-term follow-up; the sample size was too small to evaluate individual scale scores. This improvement was associated with seizure control.

Although behavioral problems are prevalent among youth with epilepsy, it is evident that the outcomes following epilepsy surgery are unclear, and reports of long-term outcomes are lacking. When improvements are reported, they are limited to specific domains, likely as a result of their multifaceted nature. In addition, half of the studies described earlier did not include a comparison group, and as such cannot determine whether change results from epilepsy surgery itself, changes in seizure frequency, or the ongoing development and maturation of the child. Change in behavioral problems may additionally require longer post-operative periods than has been used in the majority of previous studies to allow for post-surgical (or post-seizure freedom) adjustment. Only one study has attempted to examine the long-term behavioral outcomes following surgery using a broad measure of behavioral problems and a small surgical sample [10]; those results await replication in a large study of surgical and nonsurgical patients and in multiple aspects of behavioral functioning. Thus, there is a critical knowledge gap as behavior problems present a significant burden, often more so than seizures themselves, for children, families, and professionals involved in their care [11].

This study evaluated behavioral problems in a large group of surgical patients before and 4–11 years after epilepsy surgery in childhood. We evaluated, at similar time points, a comparison group of patients with intractable childhood epilepsy who did not undergo epilepsy surgery. We utilized a standardized measure of behavior that has been widely utilized in research examining behavior problems and outcomes of epilepsy surgery, the Child Behavior Checklist (CBCL) [2, 3, 6–8], and its adult counterpart, the Adult Behavior Checklist (ABCL). We have previously reported on internalizing behavioral problems and affective functioning in this cohort, finding

an advantage for seizure freedom [15]. In particular, we found that at long-term follow-up, patients with continued seizures scored significantly worse on the Anxious/Depressed subscale and marginally worse on the Withdrawn subscale of the CBCL/ABCL. In this report, we focus primarily on externalizing and related symptomology in evaluating the CBCL/ABCL Externalizing Behavior summary scale and the Somatic Complaints, Thought Problems, Attention Problems, Aggressive Behavior, Intrusive, and Rule-breaking Behavior syndrome scales. We hypothesized that improvements in behavior would be associated with seizure freedom, irrespective of surgical status.

Methods

Patients

All patients had undergone evaluations to determine potential candidacy for epilepsy surgery between 2002 and 2009 at the Hospital for Sick Children in Toronto, Ontario, Canada. Prior to surgical evaluations, seizures in all patients had failed to sufficiently improve from at least two trials of antiepileptic drugs (AEDs). Patients were deemed to be surgical candidates if a clear unilateral seizure focus could be identified and if the planned surgical site did not encroach on areas of eloquent cortex. Patients who underwent resective epilepsy surgery formed our surgical group; exclusion criteria included hemispherectomy or corpus callosotomy procedures as these individuals tend to have marked neurological impairment and these procedures may be performed for reasons other than complete seizure control, such as alleviating drop attacks. Patients who declined surgery or were not surgical candidates formed our nonsurgical comparison group; exclusion criteria included the presence of an epilepsy syndrome that would not be treated with surgery, or epilepsy associated with neurodegenerative disorders. There were 37 nonsurgical patients that participated in the study (described below), and of these individuals, 13 decided to continue pharmacological management due to improved seizure control, 5 declined surgery, 4 could not proceed with surgery as a result of eloquent cortex involvement, and 15 did not have a localizable or lateralizable epileptogenic focus; of these individuals, 6, 0, 2, and 6 were seizure-free in the 12 months preceding long-term follow-up.

We compared ratings of behavioral problems among four patient subgroups: surgical seizure-free, surgical with seizures, nonsurgical seizure-free, and nonsurgical with seizures. For the purposes of this study, seizure freedom was defined as the absence of any seizures in the year preceding the follow-up assessment.

Procedure

As part of the evaluations to determine candidacy for epilepsy surgery, patients underwent a comprehensive neuropsychological assessment, including measures of IQ and behavior (described below). Four to eleven years after surgery or evaluations for surgery, patients were contacted to participate in the current study. One-hundred and fifty-two individuals (92 surgical) met criteria for participation; 17 (6 surgical) could not be contacted, 2 (1 surgical) were deceased, and 25 (14 surgical) declined participation or did not return questionnaires. Participants were given the option of completing questionnaires over the telephone, in person or at home and returning them by mail. Overall, 108 (71 surgical and 37 nonsurgical) patients participated in the study by having data on the CBCL or ABCL at follow-up. Of these individuals, 85 (56 surgical) had CBCL data from baseline, and 93 (60 surgical) had completed IQ testing at follow-up.

This study was approved by the Research Ethics Board of the Hospital for Sick Children, and informed consent/assent was obtained from patients and/or their parents. Study data were managed and stored using REDCap [12].

Assessment

Questionnaires

At baseline, parents of patients completed the CBCL [13], and in follow-up, they again completed the CBCL if their child was 18 years old or younger, or the ABCL [14] if their child was over 18 years of age. In completing this questionnaire, parents indicated on a three-point Likert Scale, the frequency of behavioral symptoms over the past 6 months. To examine behavioral problems over time, we examined the Externalizing Behavior summary scale and a number of syndrome scales (Somatic Complaints, Thought Problems, Attention Problems, Aggressive Behavior, and Rule-breaking Behavior). In addition, at follow-up, we examined the Intrusive subscale of the ABCL; this subscale is not part of the CBCL administered at baseline. The Externalizing Behavior summary scale is a composite score of the Rule-breaking Behavior and Aggressive Behavior syndrome scales; for the ABCL, the Intrusive subscale is also included. We have previously reported on emotional functioning in the long-term following surgery in this sample using the CBCL/ABCL Internalizing Behavior, Anxious/depressed, Withdrawn, Anxiety problems, and Affective (Depression) Problems subscales [15]; for this reason, results from these subscales are not reported here. All CBCL/ABCL scores are presented as *T*-scores [mean 50, standard deviation (SD) 10], with higher scores indicative of greater problems.

Cut-off scores on the scales allow for scores to be classified as “normal”, “borderline”, or “clinical”; those in the borderline and clinical range score at ≥ 93 rd percentile. As a result of the small number of patients scoring in the borderline and clinical range, scores in these categories were grouped as “abnormal” for the purposes of this study.

Assessment of intelligence

As part of the larger study [15–18], participants also completed a comprehensive cognitive assessment, which included a measure of IQ, using the age-appropriate Wechsler Intelligence Scale [19, 20].

Statistical analyses

Participants were subdivided into four groups based on surgical status and seizure status at the long-term outcome. Demographic and epilepsy related variables of the four patient groups were compared using 2 (surgical status) $\times$ 2 (seizure status) analysis of variance (ANOVA), χ^2 , or Fisher’s exact test. In evaluating behavior outcomes, we examined the impact of surgical status, seizure status, and time using a 2 $\times$ 2 $\times$ 2 mixed factor ANOVA. Rank-transformed data were used in evaluating the Aggressive Behavior subscale as the homogeneity of variance assumption was violated; results remained similar with and without this adjustment. We also examined change at the individual level using χ^2 analyses or Fisher’s exact test, as appropriate, to compare the number of patients that had shown improvements (transition of scores from the abnormal range to the normal range), deteriorations (transition of scores from the normal range to the abnormal range), or no change from baseline to follow-up. Since baseline scores were not available for 23 patients, we evaluated behavior at follow-up among all participants using 2 (surgical status) $\times$ 2 (seizure status) ANOVA, to include these individuals in the analyses and to confirm the results of the mixed factor ANOVA. Multiple linear regressions were then used to evaluate the predictors of change from baseline to follow-up (difference scores).

Results

The mean follow-up period was 6.96 years [SD 2.25; range 4.00–11.71 years], and the mean age at surgery was 12.98 years (SD 4.34; range 4.25–18.83 years). The proportion of surgical and nonsurgical patients that were seizure-free in the 12 months preceding the follow-up was not statistically different, 54.9 and 37.8 %, respectively

$[\chi^2(1) = 2.843, p = .092]$. The most commonly used AEDs at follow-up were Carbamazepine (used by 28 participants), Clobazam (used by 21), Lamotrigine (used by 21), and Levetiracetam (used by 20). Table 1 shows the demographic and epilepsy characteristics of the four patient groups: (1) surgical seizure-free, (2) surgical with seizures, (3) nonsurgical seizure-free, and (4) nonsurgical with seizures. Comparing surgical and nonsurgical seizure-free patients, we found that surgical patients had a higher proportion of the follow-up period with controlled seizures, 92.8 vs. 57.0 %, respectively ($p = .001$). Seizure-free patients were younger at baseline, had a shorter duration of epilepsy, and had higher IQ scores at follow-up (all p values $\leq .026$). Surgical seizure-free patients had the lowest proportion of life with epilepsy ($p = .040$) and were most likely to take no AEDs at follow-up ($p < .001$). A larger proportion of nonsurgical patients with seizures had a bilateral/generalized seizure focus ($n = 5$) compared with other patient groups ($p = .011$).

There were no significant differences among demographic, epilepsy-related variables, and baseline CBCL scores between patients who did and did not participate in the study (all p values $>.05$). In addition, there were no significant differences in demographic and epilepsy-related variables between participants who completed the CBCL/ABCL at baseline and follow-up and participants who completed the questionnaires at follow-up only (all p values $>.05$).

Behavior problems over time

Mixed factor ANOVAs compared the impact of surgical status, seizure status, and time on ratings of behavioral problems. Significant main effects of time revealed that scores of Externalizing Behavior ($F_{1,81} = 4.15, p = .045$), Somatic Complaints ($F_{1,81} = 12.76, p = .001$), Thought Problems ($F_{1,79} = 4.95, p = .029$) and Attention Problems ($F_{1,80} = 10.52, p = .002$) improved from baseline to follow-up among all patient groups. Main effects of seizure-status indicated that seizure-free patients had lower (better) scores on the Externalizing Behavior ($F_{1,81} = 14.72, p < .001$), Somatic Complaints ($F_{1,81} = 7.00, p = .010$), and Aggressive Behavior ($F_{1,81} = 15.65, p < .001$) subscales. Finally, we found a significant surgical status by seizure status interaction for Attention Problems ($F_{1,80} = 5.23, p = .025$); simple effects analysis showed that among surgical patients, rating of Attention Problems was significantly lower among seizure-free patients ($M_{\text{diff}} = 8.09, p = .001$). No significant effects were found for Rule-breaking Behavior. Since there were no significant main effects of surgical status, Fig. 1 shows mean scores and standard deviation of patients with and without seizures at follow-up.

At the individual level, we compared the number of participants meeting criteria for abnormal scores. There were no significant differences between surgical and nonsurgical patients at baseline (all p values $>.20$) and follow-up (all p values $>.37$). In comparing participants with and without seizures, fewer seizure-free patients scored in the abnormal range for Attention Problems at baseline [$\chi^2(1) = 5.96, p = .015$], and in various scales at follow-up, including Externalizing Behavior [$\chi^2(1) = 9.26, p = .002$], Somatic Complaints [$\chi^2(1) = 4.78, p = .029$], and Attention Problems [$\chi^2(1) = 12.96, p < .001$; see Table 2]. Significant changes in these classifications from baseline to follow-up (improved, no change, declined) were not statistically different for surgical and nonsurgical patients (all p values $>.26$) or between patients with or without seizures (all p values $>.08$; see Table 2). Overall, 24 (28 %) patients scored in the normal range at baseline and abnormal range at follow-up in at least one domain; the opposite pattern (representing an improvement) was observed in 47 (55 %) patients.

An additional set of analyses was carried out to identify potential factors associated with the 24 individuals who scored in the normal range at baseline and abnormal range at follow-up in at least one domain. Independent samples t tests, χ^2 , or Fisher's exact test were used to compare these individuals with patients that did not show this pattern on all of the demographic and epilepsy variables listed in Table 1 as well as baseline CBCL/ABCL scores. Patients who showed deterioration over the long-term had a significantly long duration of life with seizures (10.86 vs. 14.21 years; $t_{83} = 2.096, p = .039$), a larger proportion of life with seizures (69.82 vs. 53.65 %; $t_{83} = 2424, p = .018$), and a marginally larger proportion had IQs below 70 (34.8 vs. 12.2 %; Fisher's exact test $p = .052$). Other comparisons examined were not statistically significant.

Behavior at follow-up

To take advantage of the complete sample, we evaluated behavior at follow-up with 2×2 ANOVAs comparing the impact of surgical status and seizure status. No significant main effects of surgical status were found (all p values $\geq .25$), thereby replicating the lack of a surgical status effect among those who had baseline and follow-up data. Significant effects of seizure status revealed lower (better) scores among seizure-free patients in Externalizing Behavior ($F_{1,104} = 13.50, p < .001$), Somatic Complaints ($F_{1,104} = 4.82, p = .030$), Attention Problems ($F_{1,104} = 8.97, p = .003$), Aggressive Behavior ($F_{1,104} = 12.94, p < .001$), Rule-breaking Behavior ($F_{1,104} = 7.09, p = .009$), and the Intrusive subscale ($F_{1,104} = 10.42, p = .002$; see Table 3). With the exception of a significant

Table 1 Demographic and epilepsy characteristics of the four patient groups

	Surgical seizure-free (<i>n</i> = 39)	Surgical with seizures (<i>n</i> = 32)	Nonsurgical seizure-free (<i>n</i> = 14)	Nonsurgical with seizures (<i>n</i> = 23)	<i>p</i> value
Sex, male <i>n</i> (%)	19 (49)	11 (34)	5 (36)	6 (26)	.332
Age at epilepsy onset, years, mean (SD)	6.0 (4.7)	5.5 (4.8)	4.5 (4.6)	4.8 (3.8)	≥.257
Age at baseline, years, mean (SD)	11.5 (4.5)	12.9 (4.2)	11.5 (3.7)	13.7 (2.9)	.026 ^f
Age at follow-up, years, mean (SD)	19.2 (4.2)	20.6 (4.8)	19.1 (4.1)	20.7 (4.1)	≥.103
Follow-up period, years, mean (SD) ^a	6.9 (2.3)	6.7 (2.2)	7.7 (2.3)	7.0 (2.3)	≥.284
Duration of epilepsy, years, mean (SD) ^b	6.7 (4.7)	15.1 (5.6)	10.1 (4.9)	15.9 (5.9)	<.001 ^f
Percent of life with epilepsy, mean (SD)	34.4 (21.1)	74.4 (21.2)	54.3 (26.3)	75.5 (20.4)	.040 ^h
Percent of follow-up seizure-free, mean (SD)	92.8 (18.1)	–	57.0 (30.7)	–	.001 ^g
IQ at follow-up, mean (SD)	85.7 (16.1)	74.6 (20.4)	87.0 (17.7)	77.8 (14.8)	.010 ^f
Below 70, %	12.5	39.3	8.3	19.0	.056
Not taking any AEDs at follow-up, <i>n</i> (%)	26 (66.7)	2 (6.3)	5 (35.7)	0 (0)	<.001 ⁱ
Laterality of focus, <i>n</i> (%)					.011 ^j
Right	18 (46.2)	12 (37.5)	8 (57.1)	6 (26.1)	
Left	21 (53.8)	20 (62.5)	5 (35.7)	12 (52.2)	
Bilateral	0 (0)	0 (0)	1 (7.1)	5 (21.7)	
Site of focus, <i>n</i> (%)					.159
Temporal	21 (53.8)	11 (34.4)	4 (28.6)	7 (30.4)	
Extra-temporal ^c	18 (46.2)	21 (65.6)	10 (71.4)	16 (69.6)	
Type of Resection, <i>n</i> (%)					.418
Lesionectomy	8 (20.5)	5 (15.6)	–	–	
Corticectomy	7 (17.9)	10 (31.3)	–	–	
Lobectomy	24 (61.5)	17 (53.1)	–	–	
Pathology ^d , <i>n</i> (%)					N/A
Tumour	14 (35.9)	4 (12.5)	–	–	
Malformation of cortical development	6 (15.4)	7 (21.9)	–	–	
Dual pathology	6 (15.4)	7 (21.9)	–	–	
Gliosis	6 (15.4)	3 (9.4)	–	–	
Mesial temporal sclerosis	3 (7.7)	2 (6.3)	–	–	
Vascular malformation	2 (5.1)	2 (6.3)	–	–	
Other ^e	2 (5.1)	4 (12.5)	–	–	
No discernable pathology/unknown	0 (0)	3 (9.4)	–	–	

^a Follow-up period is based on time since surgery for surgical patients, and based on time since the surgical candidacy evaluation visit for nonsurgical patients

^b Calculated as age of seizure onset and number of years seizure-free subtracted from current age

^c Includes patients who had excisions in temporal lobe as well as extratemporal regions

^d Pathology was determined by histopathological analysis in the surgical group

^e Other includes four tuberous sclerosis complex, one diffuse cytoplasmic astrocytic inclusions, one sturge–weber syndrome

^f Significant main effect of seizure status

^g Significant main effect of surgical status

^h Significant surgical status × seizure status interaction

ⁱ Significant Fisher's exact test appears to be driven by the higher proportion of surgical seizure-free patients

^j Significant Fisher's exact test appears to be driven by the higher proportion of nonsurgical patients with seizures

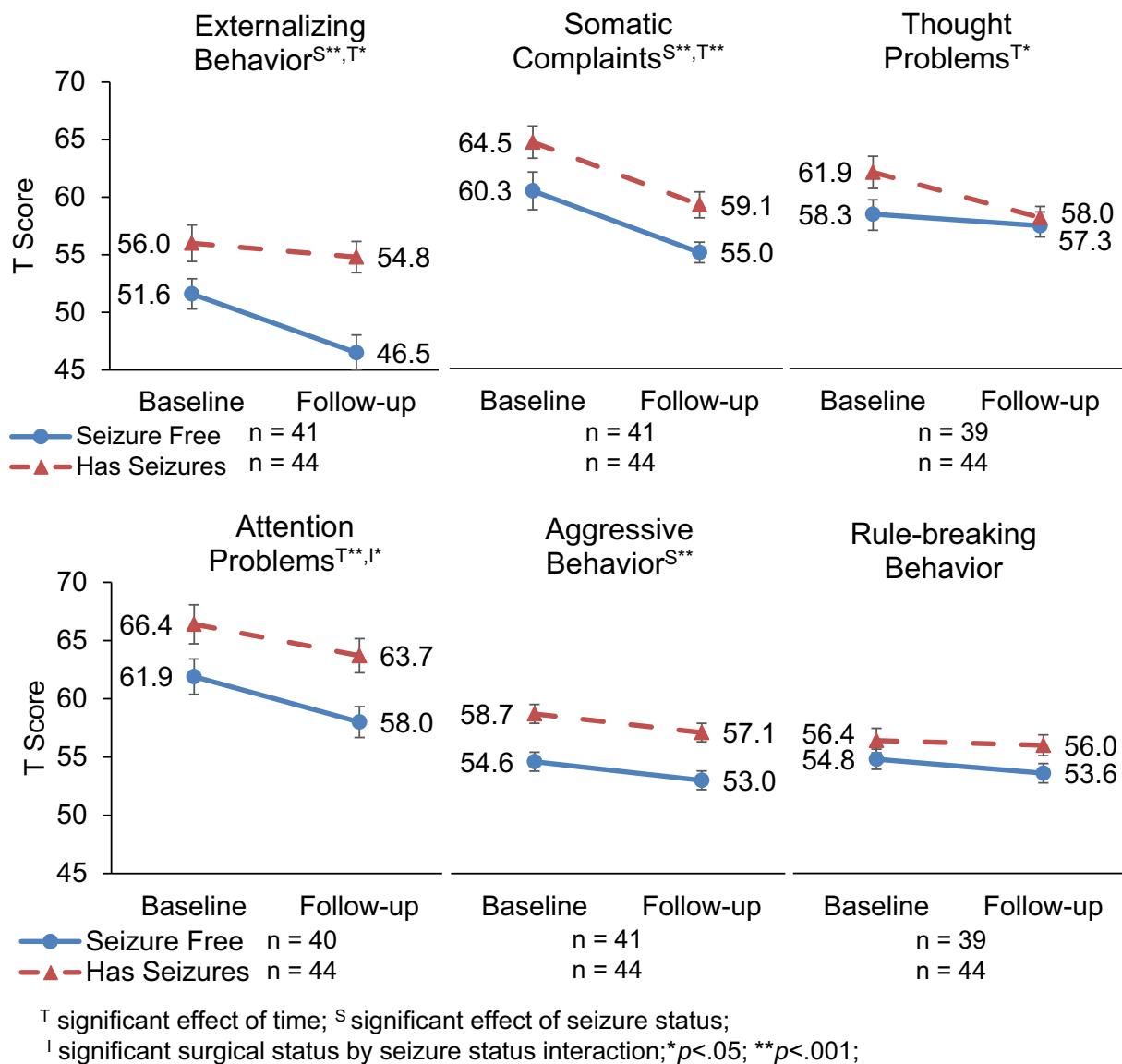


Fig. 1 Mean T-scores, standard error, and sample size of ratings of behavioral problems at baseline and follow-up

difference in Rule-breaking Behavior, these results replicate the effect of seizure status found among patients who had data at both baseline and follow-up.

An additional set of analyses was carried out to compare the impact of site (temporal vs. extratemporal) and side (left vs. right) of epileptogenic focus on ratings of behavior problems at long-term follow-up using 2×2 ANOVAs. These exploratory analyses were performed separately for patients with and without seizures; patients with bilateral/generalized epileptogenic foci were excluded from these analyses due to the small sample size ($n = 6$). Among seizure-free patients, a significant main effect of side ($F_{1,48} = 4.18$, $p = .046$) was found for the Somatic Complaints domain; patients with left epileptogenic foci

had lower (better) scores. Analyses for all other tasks were not statistically significant.

Predictors of change over time

To determine the impact of demographic or epilepsy-related variables on change in behavioral ratings from baseline to follow-up, multiple linear regressions analyses were conducted. The following variables were entered into the regression analyses: current AED use (yes vs. no), seizure status, laterality (left vs. right), site (temporal vs. extratemporal) of epileptogenic focus, age at baseline, change in IQ from baseline to follow-up, and baseline ratings of behavior. Missing variables were handled using

Table 2 Number (n) and percentage of patients with improvements or deteriorations in behavior, and the number (n) and percentage of patients who scored in the abnormal range at follow-up

	Improved		Deterioration		Abnormal range at follow-up	
	Seizure free, n (%)	Has Seizures, n (%)	Seizure free, n (%)	Has Seizures, n (%)	Seizure free, n (%)	Has Seizures, n (%)
In at least one domain	21 (51.2)	26 (59.1)	9 (22.0)	15 (34.1)	14 (26.4)**	33 (60.0)**
Externalizing Behavior	6 (14.6)	9 (20.5)	3 (7.3)	9 (20.5)	6 (11.3)**	20 (36.4)**
Somatic Complaints	11 (26.8)	12 (27.3)	3 (7.3)	5 (11.4)	5 (9.4)*	14 (25.5)*
Thought Problems	5 (12.8)	14 (31.8)	3 (7.7)	3 (6.8)	7 (13.2)	8 (14.5)
Attention Problems	9 (22.5)	10 (22.7)	4 (10.0)	4 (9.1)	8 (15.1)**	26 (47.3)**
Aggressive Behavior	2 (4.9)	7 (15.9)	2 (4.9)	6 (13.6)	4 (7.5)	11 (20.0)
Rule-breaking Behavior	1 (2.6)	4 (9.1)	3 (7.7)	2 (4.5)	3 (5.7)	6 (10.9)
Intrusive ^a	–	–	–	–	1 (3.1)	5 (12.5)

Improvement denotes a change from the abnormal range at baseline to the normal range at follow-up, whereas deterioration denotes a change from the normal range at baseline to abnormal range at follow-up

* $p < .05$, ** $p < .01$ Chi-squared analyses or Fishers exact test

^a ABCL syndrome subscale score, not available at baseline

Table 3 Mean T-scores, standard deviation (SD), and sample size (n) at follow-up

	Seizure-free, mean (SD), n	Has Seizures, mean (SD), n
Externalizing Behavior***	47.4 (9.4), 53	54.9 (10.4), 55
Somatic Complaints*	55.1 (5.5), 53	59.1 (8.5), 55
Thought Problems	56.5 (6.8), 53	58.2 (7.5), 55
Attention Problems**	57.6 (7.5), 53	64.0 (10.1), 55
Aggressive Behavior***	53.3 (5.1), 53	57.6 (7.6), 55
Rule-breaking Behavior**	53.3 (4.8), 53	56.5 (6.2), 55
Intrusive**	52.7 (4.8), 32	55.9 (6.6), 40

* $p < .05$, ** $p < .01$, *** $p < .001$

pairwise deletion. All regression models were significant ($p \leq .001$). The most consistent predictor was baseline ratings of behavior, where higher (worse) scores at baseline were associated with improvement over time for all domains (all p values $< .001$, $.318 < \beta < .842$). Seizure freedom was associated with improvement in Externalizing Behavior ($p = .009$, $\beta = .342$), and a marginally significant improvement in Aggressive Behavior ($p = .056$, $\beta = .222$). Younger age at baseline was associated with greater improvement in Externalizing Behavior ($p = .036$, $\beta = .218$). AED use, laterality, site of epileptogenic focus, and change in IQ were not significantly associated with change on any behavior outcome.

Discussion

This study examined externalizing and related behavioral problems in a large group of surgical patients before and on average 6.8 years after pediatric epilepsy surgery. Ratings

of behavioral problems in this cohort were compared against a comparison group of patients with intractable childhood epilepsy who did not undergo epilepsy surgery. Overall, we found that among all patient groups, there were significant improvements across most domains, including Externalizing Behavior, Somatic Complaints, Thought Problems, and Attention Problems; a similar improvement was not observed for Aggressive Behavior and Rule-breaking Behavior. Similar improvements among all patient groups have been reported in the short-term following surgery [8, 9] and among patients with new-onset epilepsy followed over time [21]. The observed improvements may be a result of ongoing development as the child/young adult matures and learns coping strategies, thereby changing external and internal factors with subsequent changes in ratings of behavior problems. It is also important to note that, while modest improvements were noted at the group level, a large number of patients presented with behavior problems at follow-up. In addition, 28 % of patients scored in the

normal range at baseline and in the abnormal range at follow-up in at least one domain. This finding emphasizes the importance of discussing the possibility of late emergence of behavior problems with patients and families, as patients may not be under regular review by the epilepsy center years after surgery and may not know how to access appropriate services [11]. We found that these individuals had a longer duration and proportion of life with seizures, and tended to have an IQ below 70, emphasizing the importance of attaining seizure control early and the association of low IQ with behavior problems [22].

In discerning differences among patient subgroups, we found that there were no identifiable differences in ratings of behavioral problems among surgical and nonsurgical patients. This is in keeping with the findings of the larger research project which had largely failed to identify meaningful differences between surgical and nonsurgical patients in various outcomes in the long-term, including cognition, quality of life, emotional functioning, and psychosocial and vocational outcomes [15–18]; in these domains seizure freedom was a significant predictor. Studies evaluating ratings of behavior problems in the short-term following surgery have largely found no significant differences between surgical and nonsurgical patients [3, 8, 9]. However, in contrast to several short-term outcome studies [6, 8, 9], we found meaningful differences associated with seizure freedom, in that, seizure-free patients had fewer behavioral problems across most domains. In terms of Attention Problems, the advantage of seizure freedom was present among surgical patients only, possibly related to the longer duration of seizure freedom and fewer AEDs used by surgical patients in this study. These results were reflected at the individual level, where fewer seizure-free patients scored in the abnormal range for the summary score and some syndrome scores at follow-up. Although some studies have also reported improvements associated with seizure control, they have utilized small samples, short follow-up periods (3–17 months), and generally lacked controls groups [2, 7]; thus, our findings provide a stronger confirmation of these effects.

We were limited in investigating the mechanism through which seizure freedom was associated with fewer behavior problems. There are likely a multitude of factors that subsequently lead to fewer behavioral problems, many of which may be brought upon by seizure control. Beyond the influence on internal factors, it may be that seizure freedom allows for a restructuring of external factors, such as family functioning and parental mental health. These external variables are known to have a negative impact on patients' behavior problems [21, 23], and have been shown to subside in association with seizure control [24, 25]. Family factors have also been shown to negatively impact behavior problems independent of demographic,

neuropsychological, and seizure variables, suggesting that targeting parenting behaviors and family environment may further aid in addressing patient behavioral problems [21]. Patient and parental social support and life stress may also impact behavior problems. Our study may have identified widespread improvements associated with seizure freedom, because patients and their families may require time to distance themselves from the worry of seizures and to engage more fully in activities and peer interactions [9].

We also investigated the association of demographic and epilepsy-related variables with behavioral problems. Recent studies have shown a possible association between the location of epileptogenic foci and behavior outcomes following surgery in the short-term [6, 8]. We were largely unable to replicate these findings in the long-term, likely as a result of the heterogeneity of our sample and smaller sample size when divided into subgroups. We found that among seizure-free patients, those with the left hemisphere epileptic foci had lower (better) scores of Somatic Complaints at follow-up in comparison with patients with the right-sided foci. In regression analyses, side and site of epileptogenic focus were also not associated with change in behavior problems over time. In contrast, seizure-freedom was associated with improved Externalizing Behavior. Younger age at baseline was associated with improved Externalizing Behavior, indicating the advantage of relative youth in making adaptive gains at the rate of their healthy peers [8]. The most consistent predictor of improved behavior problems across all subscales was greater problems at baseline, which may reflect regression toward the mean or the overall trend of improved behavior problems over time.

Strengths and limitations

This study reports on selected behavioral problems, and is the first to evaluate these measures in the long-term (four or more years) following pediatric epilepsy surgery. The use of a large heterogeneous cohort, in terms of epileptogenic foci, allows for a representative sample of children who are considered for epilepsy surgery. However, such a heterogeneous sample makes it difficult to explore potential differences associated with site and laterality of seizure foci and areas of surgical resection. We were also limited in evaluating family and environmental factors, which may have influenced behavior problems. In maintaining consistency with baseline measures, we were unable to corroborate parental reports of behavior problems with patient self reports. Finally, our sample was comprised of surgical and nonsurgical patients that were well matched on a number of demographic and seizure-related variables at baseline, and similar comparison groups are frequently utilized in evaluating surgical outcomes [8, 9]. In addition, previous investigations have shown that these groups do not significantly

differ in multiple cognitive and psychosocial domains [26]. At follow-up, 55 % of the surgical patients and 38 % of nonsurgical patients were seizure-free in the 12 months preceding the follow-up, similar to other reports [27, 28]. A nonsurgical comparison group allows us to discern whether observed outcomes were the result of epilepsy surgery, seizure freedom, or the ongoing development of the child.

Conclusion

In evaluating behavior problems in a large cohort of surgical and nonsurgical patients with intractable childhood epilepsy, we found widespread modest behavioral improvements in the long-term. Nonetheless, a large proportion of patients scored in the abnormal range of behavior problems at follow-up, and behavior problems emerged at follow-up in nearly 30 % of patients. No identifiable differences between surgical and nonsurgical patients were found; however, it is important to note that, surgical patients achieved seizure freedom sooner and used fewer AEDs at follow-up. Findings were largely related to seizure-status, in that, seizure-free patients presented with fewer behavior problems at follow-up. We were unable to evaluate whether the effects of seizure freedom in this study were associated with improved family and environmental variables, and whether improvements in family and environmental variables reduce behavior problems. It is essential that behavior and mental health problems in children with epilepsy be monitored closely by physicians, and families be educated on the risks and treatment options of behavioral problems. These findings are important in understanding the long-term outcomes of pediatric epilepsy surgery and may assist health care professionals, families, and patients considering surgery.

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Compliance with ethical standards

Conflicts of interest Neither of the authors has any conflicts of interest to disclose.

Ethical standards The study was approved by the Research Ethics Board of the Hospital for Sick Children, and performed in accordance with ethical standards.

Informed consent Informed consent/assent was obtained from patients and/or their parents prior to study participation.

References

- Ott D, Siddarth P, Gurbani S, Koh S, Tournay A, Shields WD, Caplan R (2003) Behavioral disorders in pediatric epilepsy: unmet psychiatric need. *Epilepsia* 44:591–597
- Lendt M, Helmstaedter C, Kuczaty S, Schramm J, Elger CE (2000) Behavioural disorders in children with epilepsy: early improvement after surgery. *J Neurol Neurosurg Psychiatry* 69:739–744
- Smith ML, Elliott IM, Lach L (2004) Cognitive, psychosocial, and family function one year after pediatric epilepsy surgery. *Epilepsia* 45:650–660. doi:10.1111/j.0013-9580.2004.21903.x
- McLellan A, Davies S, Heyman I, Harding B, Harkness W, Taylor D, Neville BG, Cross JH (2005) Psychopathology in children with epilepsy before and after temporal lobe resection. *Dev Med Child Neurol* 47:666–672. doi:10.1017/S0012162205001362
- Titus JB, Lee A, Kasasbeh A, Thio LL, Stephenson J, Steger-May K, Limbrick DD Jr, Smyth MD (2013) Health-related quality of life before and after pediatric epilepsy surgery: the influence of seizure outcome on changes in physical functioning and social functioning. *Epilepsy Behav* 27:477–483. doi:10.1016/j.yebeh.2013.03.003
- Andresen EN, Ramirez MJ, Kim KH, Dorfman AB, Haut JS, Klaas PA, Jehi LE, Shea K, Bingaman WE, Busch RM (2014) Effects of surgical side and site on mood and behavior outcome in children with pharmacoresistant epilepsy. *Front Neurol* 5:18. doi:10.3389/fneur.2014.00018
- Williams J, Griebel ML, Sharp GB, Boop FA (1998) Cognition and behavior after temporal lobectomy in pediatric patients with intractable epilepsy. *Pediatr Neurol* 19:189–194
- Law N, Kerr EN, Smith ML (2015) Evaluation of behavioural outcomes in children 1 year after epilepsy surgery. *Epilepsia*. doi:10.1111/epi.13126
- Elliott IM, Lach L, Kadis DS, Smith ML (2008) Psychosocial outcomes in children 2 years after epilepsy surgery: has anything changed? *Epilepsia* 49:634–641. doi:10.1111/j.1528-1167.2007.01498.x
- Hannan S, Cross JH, Scott RC, Harkness W, Heyman I (2009) The effects of epilepsy surgery on emotions, behavior, and psychosocial impairment in children and adolescents with drug-resistant epilepsy: a prospective study. *Epilepsy Behav* 15:318–324. doi:10.1016/j.yebeh.2009.04.007
- McLellan A (2015) Long-term psychiatric and behavior outcomes in children following epilepsy surgery. In: Malmgren K, Baxendale S, Cross JH (eds) *Long-Term Outcomes of Epilepsy Surgery in Adults and Children*. Springer International Publishing. doi:10.1007/978-3-319-17783-0_8
- Harris PA, Taylor R, Thielke R, Payne J, Gonzalez N, Conde JG (2009) Research electronic data capture (REDCap)—A metadata-driven methodology and workflow process for providing translational research informatics support. *J Biomed Inform* 42:377–381
- Achenbach TM, Rescorla LA (2001) *Manual for the ASEBA School-Age Forms and Profiles*. University of Vermont, Research Center for Children, Youth & Families, Burlington, VT
- Achenbach TM, Rescorla LA (2003) *Manual for the ASEBA Adult Forms and Profiles*. University of Vermont, Research Center for Children, Youth & Families, Burlington, VT

15. Tavares TP, Puka K, Smith ML (2015) Emotional functioning: long-term outcomes after pediatric epilepsy surgery. *Epilepsia* 56:745–753. doi:[10.1111/epi.12956](https://doi.org/10.1111/epi.12956)
16. Puka K, Tavares TP, Smith ML (2015) Development of intelligence 4 to 11 years after paediatric epilepsy surgery. *J Neuropsychol*. doi:[10.1111/jnp.12081](https://doi.org/10.1111/jnp.12081)
17. Puka K, Smith ML (2015) Predictors of long-term quality of life after pediatric epilepsy surgery. *Epilepsia* 56:873–881. doi:[10.1111/epi.13004](https://doi.org/10.1111/epi.13004)
18. Puka K, Smith ML (2016) Where are they now? Psychosocial, educational and vocational outcomes after epilepsy surgery in childhood. *Epilepsia*. doi:[10.1111/epi.13327](https://doi.org/10.1111/epi.13327)
19. PsychologicalCorporation (1997) WAIS-III WMS-III technical manual. The Psychological Corporation, San Antonio
20. Wechsler D (2003) Manual for Wechsler Intelligence Scale for Children, 4th edn. The Psychological Corporation, San Antonio
21. Austin JK, Perkins SM, Johnson CS, Fastenau PS, Byars AW, deGrauw TJ, Dunn DW (2011) Behavior problems in children at time of first recognized seizure and changes over the following 3 years. *Epilepsy Behav* 21:373–381. doi:[10.1016/j.yebeh.2011.05.028](https://doi.org/10.1016/j.yebeh.2011.05.028)
22. Buelow JM, Austin JK, Perkins SM, Shen J, Dunn DW, Fastenau PS (2003) Behavior and mental health problems in children with epilepsy and low IQ. *Dev Med Child Neurol* 45:683–692
23. Ferro MA, Speechley KN (2009) Depressive symptoms among mothers of children with epilepsy: a review of prevalence, associated factors, and impact on children. *Epilepsia* 50:2344–2354. doi:[10.1111/j.1528-1167.2009.02276.x](https://doi.org/10.1111/j.1528-1167.2009.02276.x)
24. Lv R, Wu L, Jin L, Lu Q, Wang M, Qu Y, Liu H (2009) Depression, anxiety and quality of life in parents of children with epilepsy. *Acta Neurol Scand* 120:335–341. doi:[10.1111/j.1600-0404.2009.01184.x](https://doi.org/10.1111/j.1600-0404.2009.01184.x)
25. Karakis I, Montouris GD, Piperidou C, Luciano MS, Meador KJ, Cole AJ (2013) The effect of epilepsy surgery on caregiver quality of life. *Epilepsy Res* 107:181–189. doi:[10.1016/j.eplepsyres.2013.08.006](https://doi.org/10.1016/j.eplepsyres.2013.08.006)
26. Smith ML, Elliott IM, Lach L (2002) Cognitive skills in children with intractable epilepsy: comparison of surgical and nonsurgical candidates. *Epilepsia* 43:631–637
27. Edelvik A, Rydenhag B, Olsson I, Flink R, Kumlien E, Kallen K, Malmgren K (2013) Long-term outcomes of epilepsy surgery in Sweden: a national prospective and longitudinal study. *Neurology* 81:1244–1251. doi:[10.1212/WNL.0b013e3182a6ca7b](https://doi.org/10.1212/WNL.0b013e3182a6ca7b)
28. Skirrow C, Cross JH, Cormack F, Harkness W, Vargha-Khadem F, Baldeweg T (2011) Long-term intellectual outcome after temporal lobe surgery in childhood. *Neurology* 76:1330–1337. doi:[10.1212/WNL.0b013e31821527f0](https://doi.org/10.1212/WNL.0b013e31821527f0)